Assignment 5 Solution

Q1. When multiplying two triangular fuzzy numbers, what is a possible outcome?

(a) Always a triangular fuzzy number

(b) A trapezoidal fuzzy number

(c) A crisp number

(d) Not guaranteed to maintain triangular shape

Solution: (d)

Explanation: Multiplication can distort the triangular shape, potentially violating convexity or normality.

Q2. Which property ensures that a fuzzy set's α-cuts are intervals?

(a) Normality

(b) Convexity

(c) Symmetry

(d) Completeness

Solution: (b)

Explanation: Convexity guarantees that α-cuts are intervals, a key requirement for fuzzy numbers.

Q3. For continuous fuzzy numbers, division involves:

(a) Summing membership values

(b) Integrating over the domain

(c) Subtracting supports

(d) Multiplying α-cuts

(e) None of the above

Solution: (e)

Explanation: For continuous fuzzy numbers, division is performed using α-cut arithmetic, meaning the division operation is carried out at each α-cut level. Given two fuzzy numbers A and B (where, B does not contain zero), their division is computed as:

$$\boxed{C = \sum_{X} \mu_C(x^C)/x^C = \sum_{X} \mu_{A \div B}(x^C)/x^C = \sum_{X} \max_{x^A, x^B} [\, \mu_A(x^A) \wedge \mu_B(x^B)]/x^C}$$

where $x^C = x^A \div x^B; \forall\, x^A, x^B, x^C \in X$

This means:

- Membership values are not summed (eliminating option a).
- Integration is not required for division (eliminating option b).
- Subtracting supports does not define division in fuzzy arithmetic (eliminating option c).
- Multiplying α-cuts is incorrect, as division follows interval arithmetic at α-cuts (eliminating option d).

Since none of the given choices correctly describe the division of fuzzy numbers, the correct answer is (e) None of the above.

Q4. Dividing two fuzzy numbers might result in:

(a) A non-convex set

(b) A non-normal set

(c) Both (a) and (b)

(d) Always a valid fuzzy number

Solution: (c)

Explanation: Division can violate both convexity and normality, disqualifying the result as a fuzzy number.

Q5. Why is the universe of discourse critical in fuzzy arithmetic?

(a) To enforce binary logic compatibility

(b) To limit operations to meaningful ranges

(c) To speed up computations

(d) To ensure symmetry

Solution: (b)

Explanation: The universe of discourse (UoD) is the complete range of values that a variable can take. In fuzzy arithmetic, the UoD is crucial because it defines the domain within which fuzzy sets and their operations are valid.Fuzzy logic operates on degrees of truth rather than binary (crisp) logic, so enforcing binary compatibility is not the purpose of UoD.The UoD ensures that fuzzy arithmetic operations remain within a logically and contextually appropriate range.While a well-defined UoD may improve efficiency, its primary role is not to speed up computations but to maintain meaningful calculations.The UoD does not necessarily ensure symmetry in fuzzy arithmetic.Thus, the correct answer is (b), as the universe of discourse primarily limits fuzzy operations to meaningful ranges.

Q6. The complement of a fuzzy set A with membership $\mu_A(x)$ = 0.7 is:

(a) 0

(b) 1/0.7

(c) 1 - 0.7

(d) 0.7^2

Solution: (c)

Explanation: The basic fuzzy complement is defined as: 1-$\mu_A(x)$.

Q7. If Sugeno's complement parameter λ approaches infinity, what happens?

(a) Complement approaches 0

(b) Complement approaches 1

(c) Complement equals 1 - μ

(d) Complement becomes undefined

Solution: (a)

Explanation: For large λ, $N(\mu) = \frac{1-\mu}{1+\lambda\mu}$ approaches 0 (if μ> 0).

Q8. In Yager's complement, if w=2, the complement of μ=0.5 is:

(a) $\sqrt{1-0.5^2}$

(b) $(1-0.5)^{\frac{1}{2}}$

(c) $1-0.5$

(d) $\sqrt{0.5}$

Solution: (a)

Explanation: Yager's formula:$N(\mu) = (1-\mu^w)^{\frac{1}{w}}$. For w=2:$\sqrt{1-0.5^2} = \sqrt{0.75}$.

Q9. Which T-norm produces the highest membership value for intersection between$\mu_A(x)$=0.6 and $\mu_B(x)$=0.8?

(a) Minimum

(b) Algebraic product

(c) Bounded product

(d) Drastic product

Solution: (a)

Explanation: Minimum (0.6) is the largest T-norm value here.

Q10. Which axiom is unique to T-norms but not S-norms?

(a) $T(1, a) = a$

(b) $S(0. a) = a$

(c) Associativity

(d) Non-decreasing

Solution: (a)

Explanation: The boundary condition $T(1, a) = a$ ensures compatibility with crisp logic for T-norms.